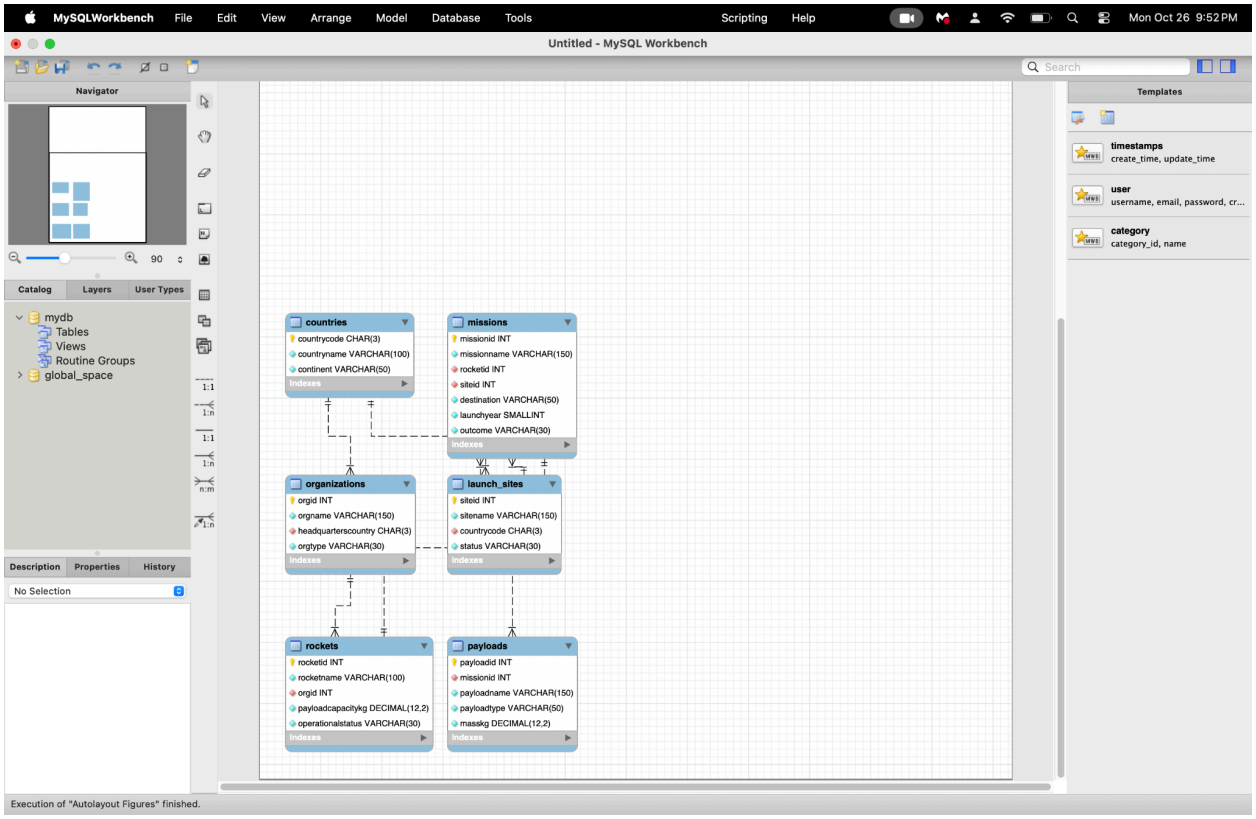
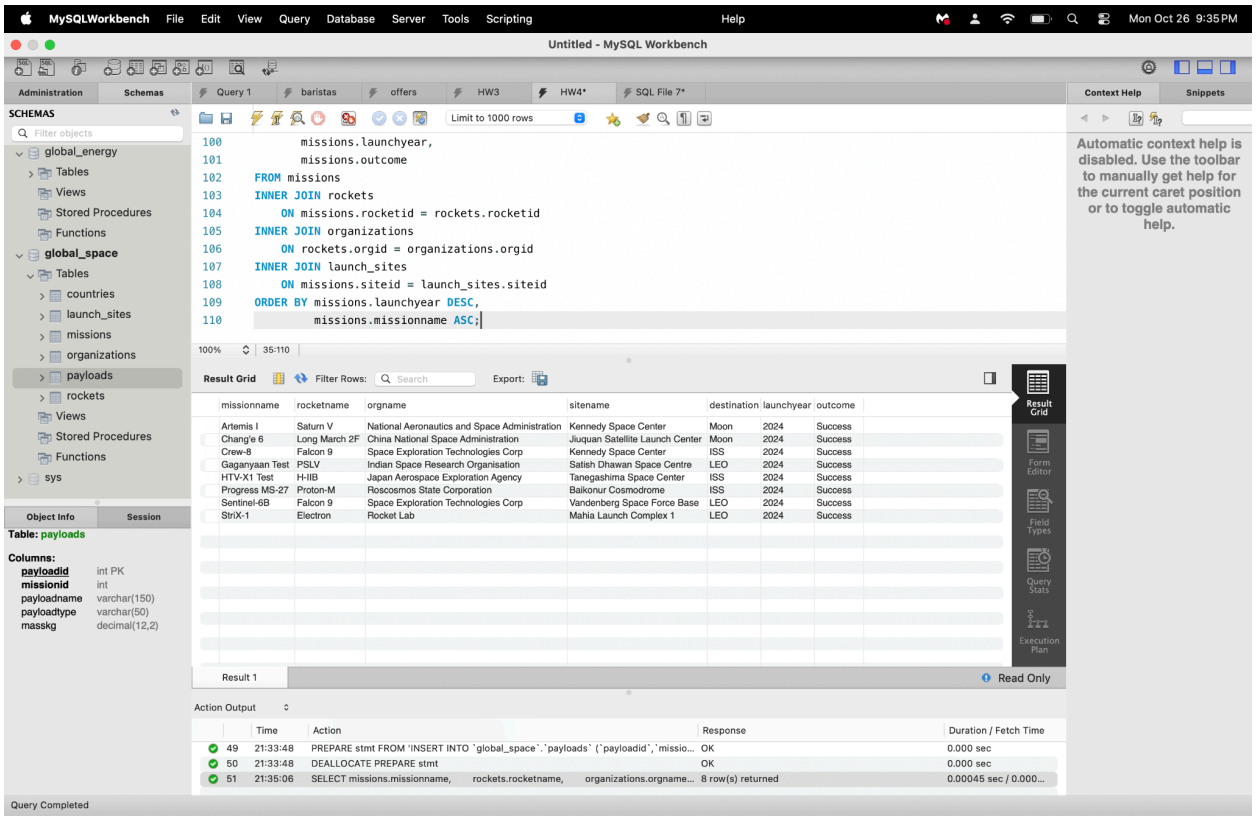
Najeeb Phillips
Assignment: HW4
Global Space Mission Database



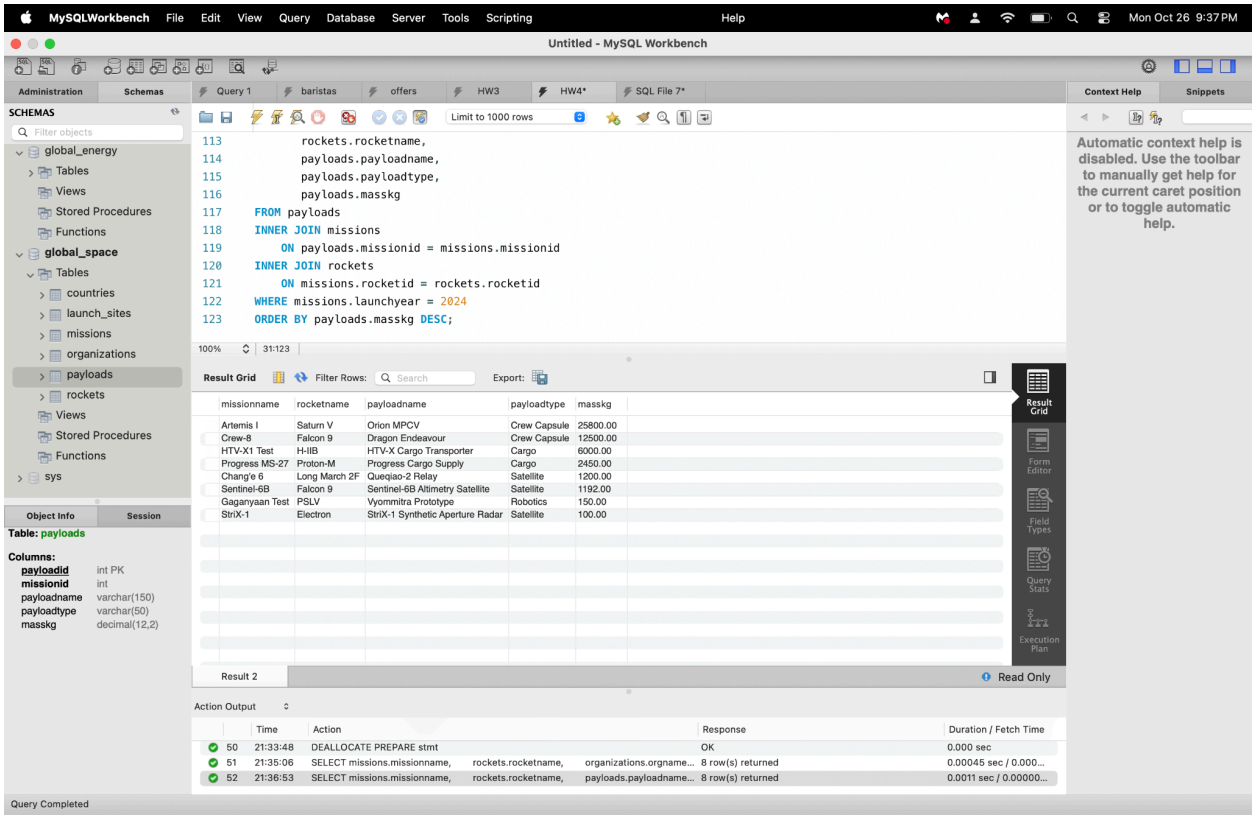
Query 1:All Space Missions

```
SELECT missions.missionname,
       rockets.rocketname,
       organizations.orgname,
       launch_sites.sitename,
       missions.destination,
       missions.launchyear,
       missions.outcome
FROM missions
INNER JOIN rockets
   ON missions.rocketid = rockets.rocketid
INNER JOIN organizations
   ON rockets.orgid = organizations.orgid
INNER JOIN launch_sites
   ON missions.siteid = launch_sites.siteid
ORDER BY missions.launchyear DESC,
        missions.missionname ASC;
```



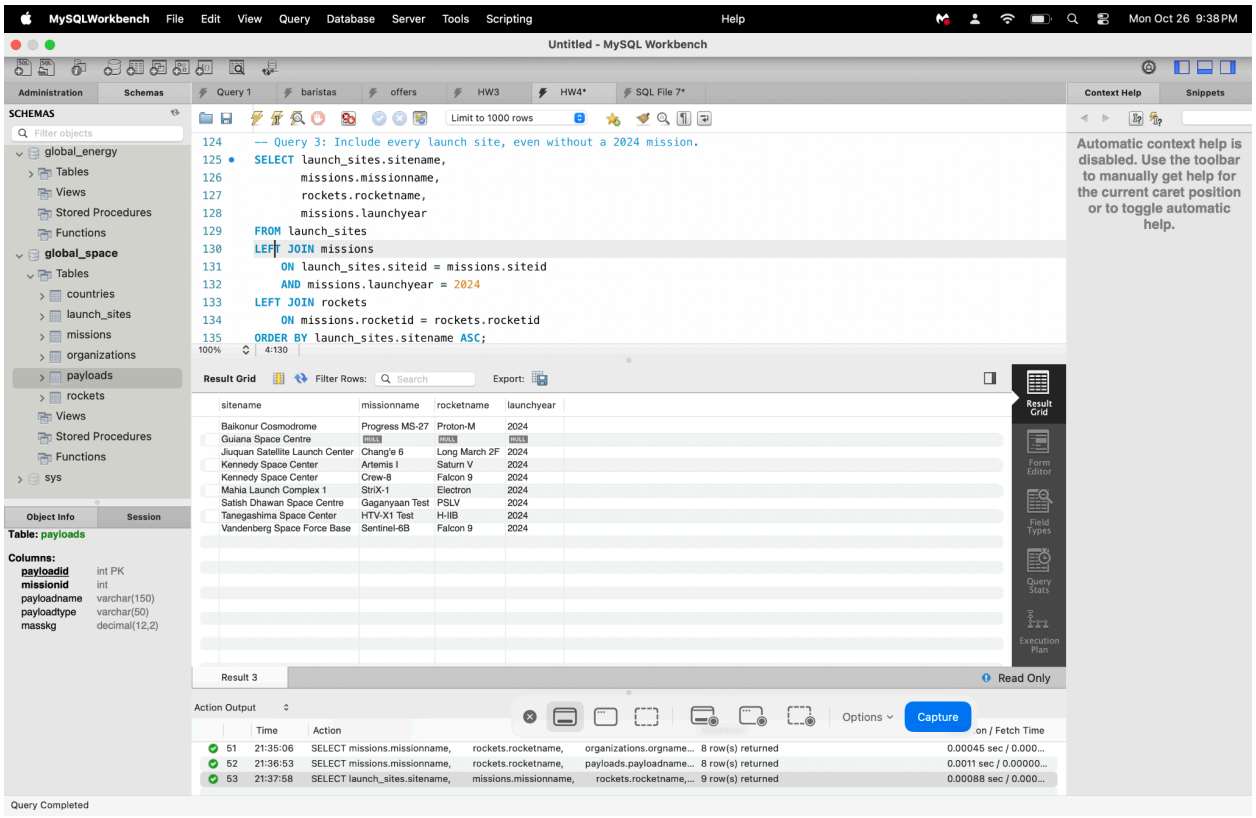
Query 2: Payloads Launched in 2024

```
SELECT missions.missionname,
       rockets.rocketname,
       payloads.payloadname,
       payloads.payloadtype,
       payloads.masskg
FROM payloads
INNER JOIN missions
   ON payloads.missionid = missions.missionid
INNER JOIN rockets
   ON missions.rocketid = rockets.rocketid
WHERE missions.launchyear = 2024
ORDER BY payloads.masskg DESC;
```



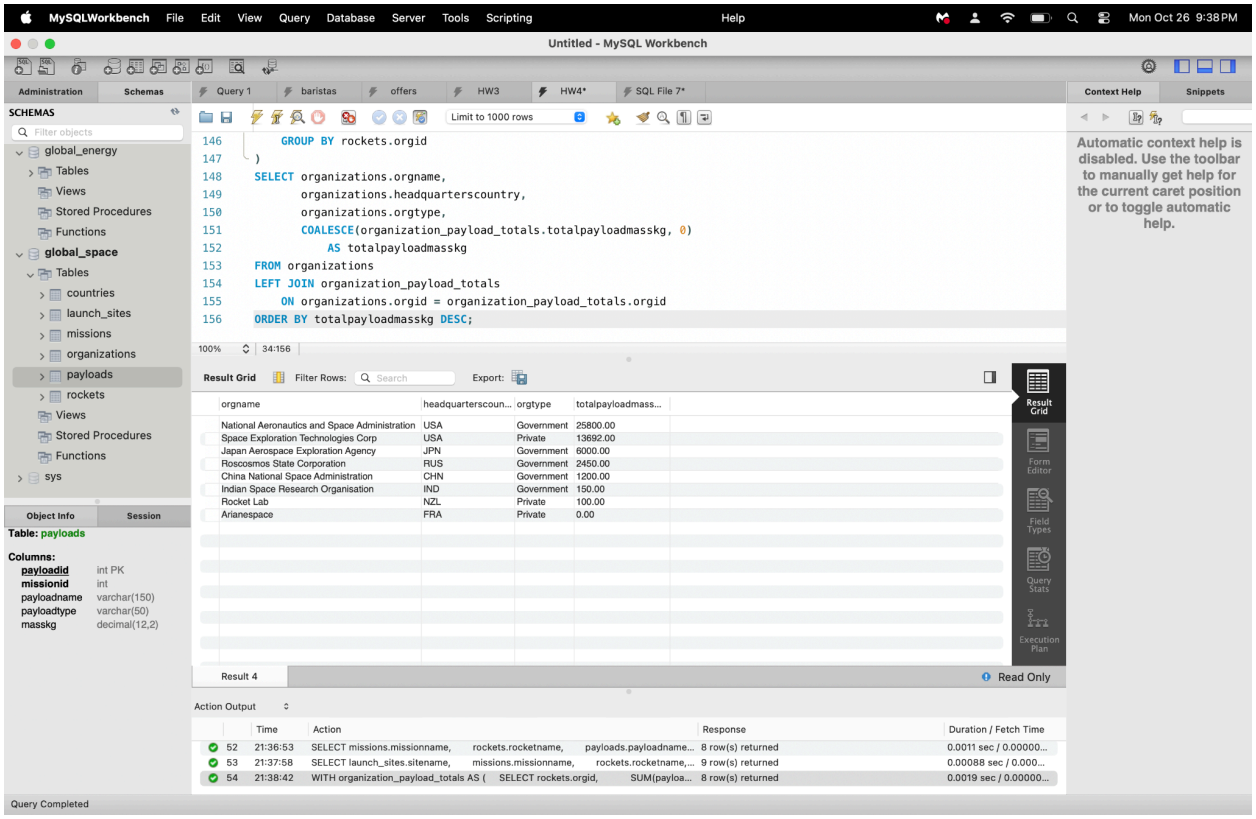
Query 3: All launch Sites and Their 2024 Missions

```
SELECT launch_sites.sitename,
       missions.missionname,
       rockets.rocketname,
       missions.launchyear
FROM launch_sites
LEFT JOIN missions
   ON launch_sites.siteid = missions.siteid
   AND missions.launchyear = 2024
LEFT JOIN rockets
   ON missions.rocketid = rockets.rocketid
ORDER BY launch_sites.sitename ASC;
```



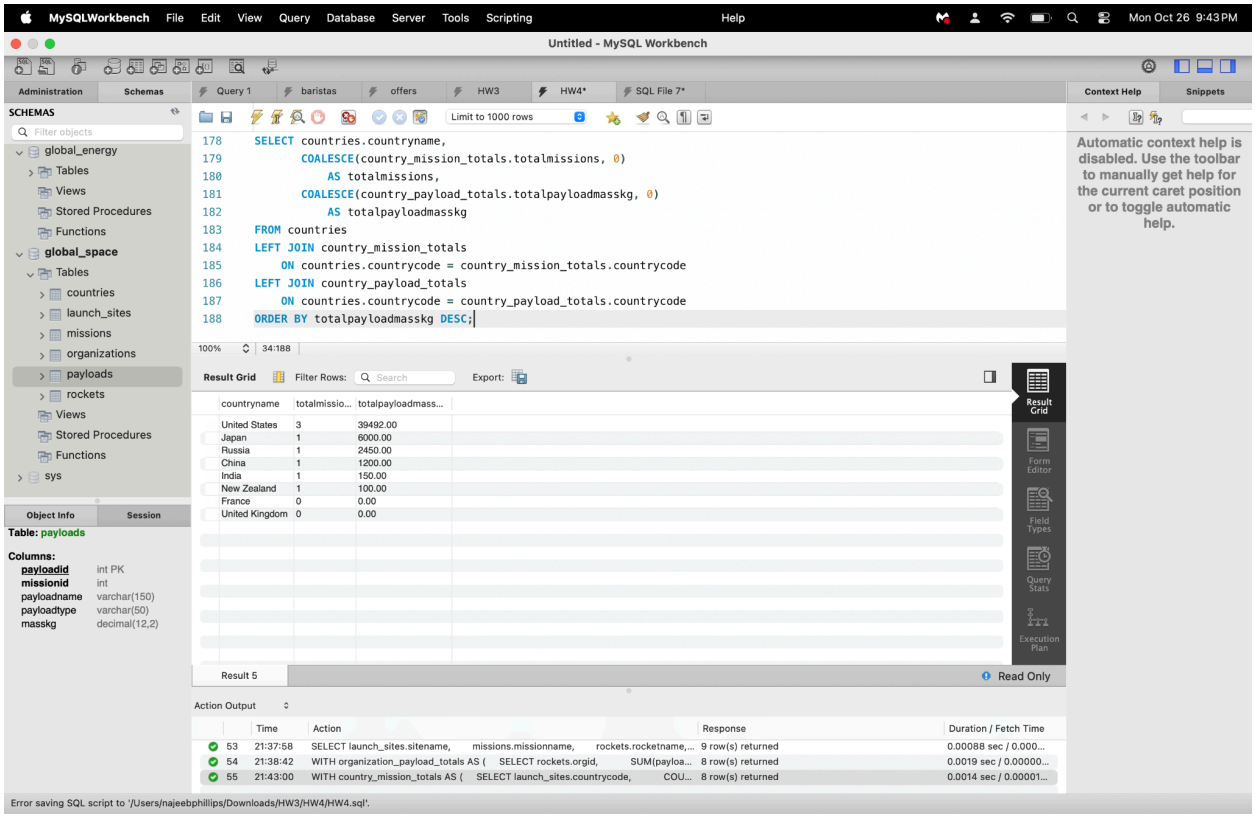
Query 4: Total payload Mass by Organization

```
WITH organization_payload_totals AS (
    SELECT rockets.orgid,
           SUM(payloads.masskg) AS totalpayloadmasskg
    FROM rockets
    INNER JOIN missions
        ON rockets.rocketid = missions.rocketid
    INNER JOIN payloads
        ON missions.missionid = payloads.missionid
    GROUP BY rockets.orgid
)
SELECT organizations.orgname,
       organizations.headquarterscountry,
       organizations.orgtype,
       COALESCE(organization_payload_totals.totalpayloadmasskg, 0)
           AS totalpayloadmasskg
FROM organizations
LEFT JOIN organization_payload_totals
    ON organizations.orgid = organization_payload_totals.orgid
ORDER BY totalpayloadmasskg DESC;
```



Query 5: Total Missions and Payload Mass by Country

```
WITH country_mission_totals AS (
   SELECT launch_sites.countrycode,
         COUNT(missions.missionid) AS totalmissions
   FROM launch_sites
   INNER JOIN missions
      ON launch_sites.siteid = missions.siteid
   GROUP BY launch_sites.countrycode
),
country_payload_totals AS (
   SELECT launch_sites.countrycode,
         SUM(payloads.masskg) AS totalpayloadmasskg
   FROM launch_sites
   INNER JOIN missions
      ON launch_sites.siteid = missions.siteid
   INNER JOIN payloads
      ON missions.missionid = payloads.missionid
   GROUP BY launch_sites.countrycode
)
SELECT countries.countryname,
      COALESCE(country_mission_totals.totalmissions, 0)
         AS totalmissions,
      COALESCE(country_payload_totals.totalpayloadmasskg, 0)
         AS totalpayloadmasskg
FROM countries
LEFT JOIN country_mission_totals
   ON countries.countrycode = country_mission_totals.countrycode
LEFT JOIN country_payload_totals
   ON countries.countrycode = country_payload_totals.countrycode
ORDER BY totalpayloadmasskg DESC;
```


MySQLWorkbench File Edit View Query Database Server Tools Scripting Help Mon Oct 26 9:43 PM
Untitled - MySQL Workbench
Administration Schemas
SCHEMAS
Filter objects
global_energy
Tables
Views
Stored Procedures
Functions
global_space
Tables
countries
launch_sites
missions
organizations
payloads
rockets
Views
Stored Procedures
Functions
sys
Object Info Session
Table: payloads
Columns:
payloadid int PK
missionid int
payloadname varchar(150)
payloadtype varchar(50)
masskg decimal(12,2)
Query 1 baristas offers HW3 HW4* SQL File 7*
Limit to 1000 rows
178 SELECT countries.countryname,
179 COALESCE(country_mission_totals.totalmissions, 0)
180 AS totalmissions,
181 COALESCE(country_payload_totals.totalpayloadmasskg, 0)
182 AS totalpayloadmasskg
183 FROM countries
184 LEFT JOIN country_mission_totals
185 ON countries.countrycode = country_mission_totals.countrycode
186 LEFT JOIN country_payload_totals
187 ON countries.countrycode = country_payload_totals.countrycode
188 ORDER BY totalpayloadmasskg DESC;
100% 34:188
Result Grid Filter Rows: Search Export:
countryname totalmissio... totalpayloadmass...
United States 3 39492.00
Japan 1 6000.00
Russia 1 2450.00
China 1 1200.00
India 1 150.00
New Zealand 1 100.00
France 0 0.00
United Kingdom 0 0.00
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
Result 5
Read Only
Action Output
Time Action Response Duration / Fetch Time
53 21:37:58 SELECT launch_sites.sitename, missions.missionname, rockets.rocketname,... 9 row(s) returned 0.00088 sec / 0.000...
54 21:38:42 WITH organization_payload_totals AS (SELECT rockets.orgid, SUM(payloa... 8 row(s) returned 0.0019 sec / 0.00000...
55 21:43:00 WITH country_mission_totals AS (SELECT launch_sites.countrycode, COU... 8 row(s) returned 0.0014 sec / 0.00001...
Context Help Snippets
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Error saving SQL script to '/Users/najeebphillips/Downloads/HW3/HW4/HW4.sql'.